

Sri Sivasubramaniya Nadar College of Engineering, Chennai
(An Autonomous Institution affiliated to Anna University)

Degree & Branch	B.E Computer Science & Engineering	Semester	VI
Subject Code & Name	UCS2612 – Machine Learning Laboratory		
Academic Year	2025–2026 (Even)	Batch	2023–2027

Experiment 6

Decision Tree and Random Forest: A Comparative Classification Study

Objective

- To implement a **Decision Tree** classifier.
- To extend the Decision Tree into a **Random Forest ensemble model**.
- To study the impact of hyperparameters on overfitting and generalization.
- To select **optimal hyperparameters using 5-Fold Cross-Validation**.
- To compare single-tree and ensemble-tree models.

Dataset

Wisconsin Diagnostic Breast Cancer Dataset

- Total samples: 569
- Features: 30 numerical attributes
- Target classes: Malignant (M) and Benign (B)

Dataset Link: <https://archive.ics.uci.edu>

Hyperparameters to be Explored

Decision Tree

- `criterion`
- `max_depth`
- `min_samples_split`
- `min_samples_leaf`

Random Forest

- `n_estimators`
- `max_depth`
- `max_features`
- `bootstrap`

Graphs

Decison Tree

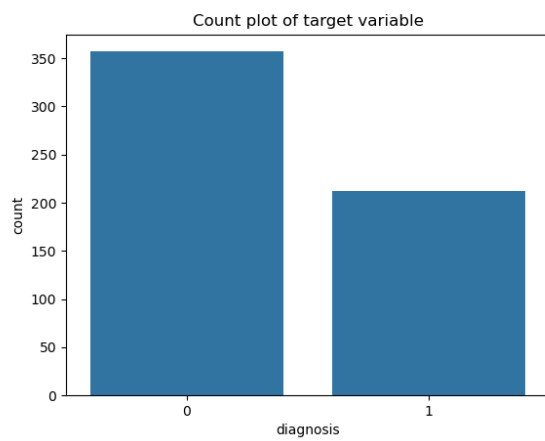


Figure 1: Count plot of target variable

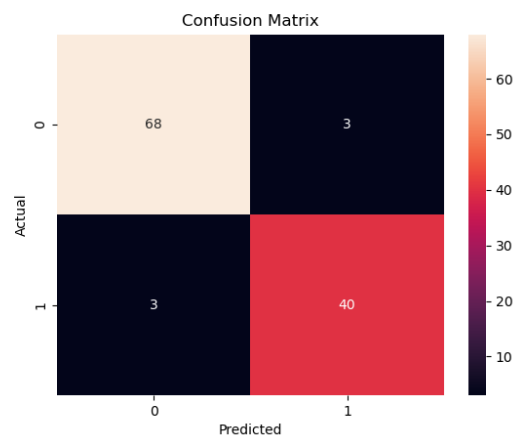


Figure 2: Confusion Matrix

Figure 3: Overall figure caption

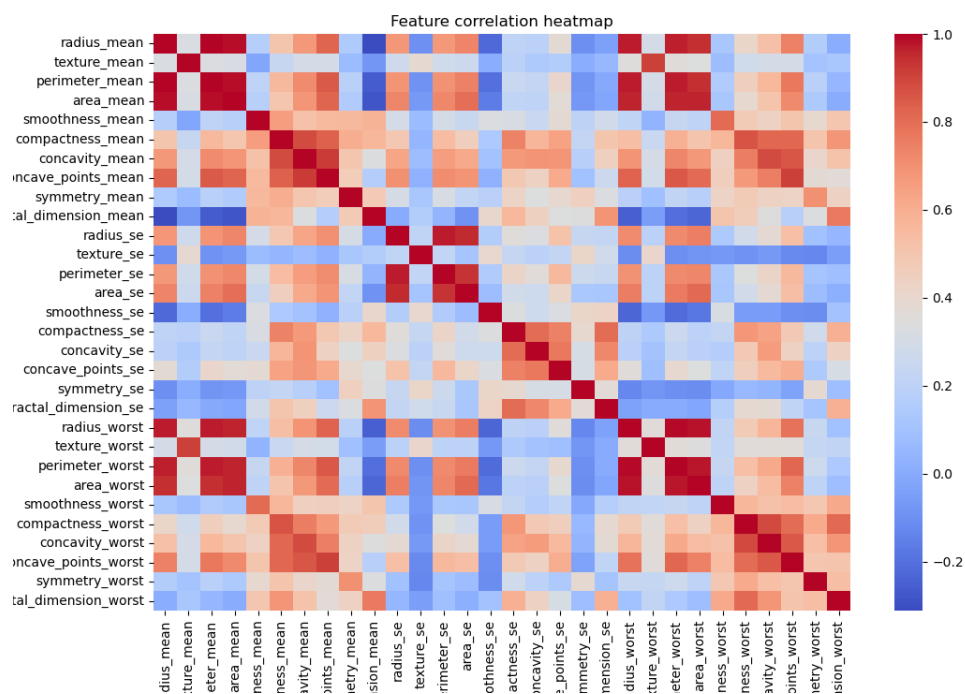


Figure 4: Feature Correlation heatmap

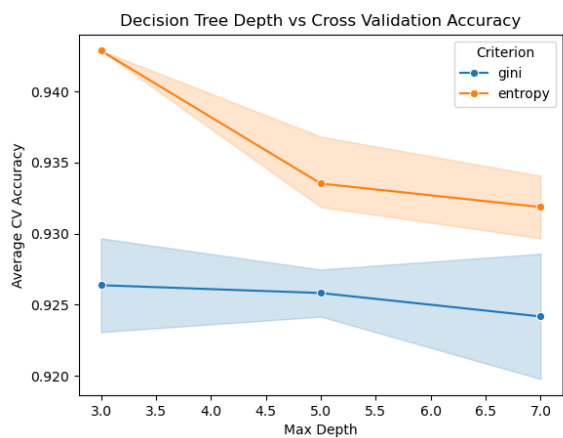


Figure 5: Decision Tree Depth vs Cross Validation Accuracy

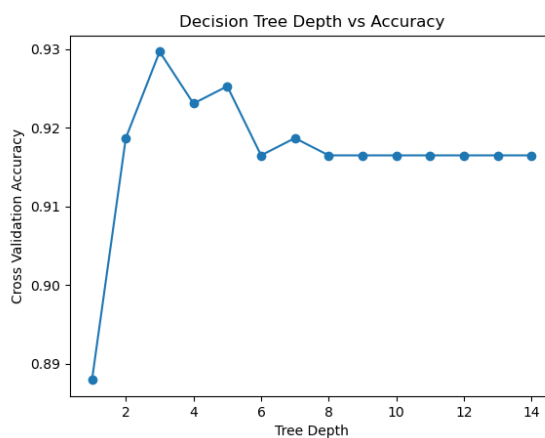


Figure 6: Decision Tree Depth vs Accuracy

Figure 7: Overall figure caption

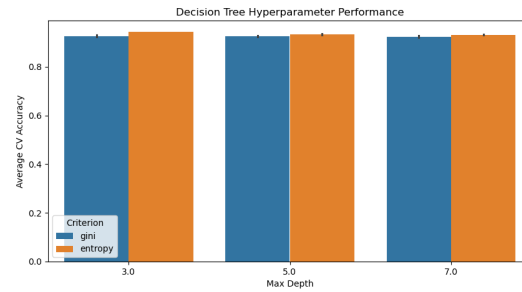


Figure 8: Decision Tree Hyperparameter Performance

Random Forest

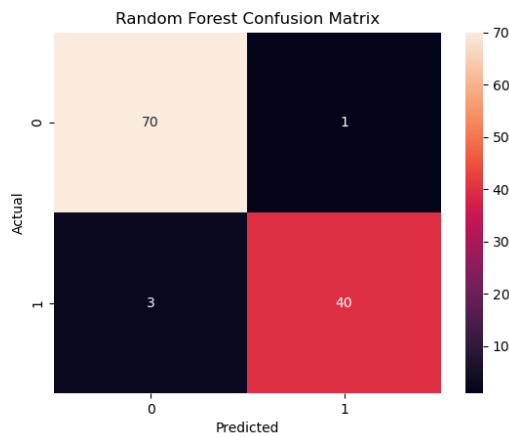


Figure 9: Random Forest Confusion Matrix

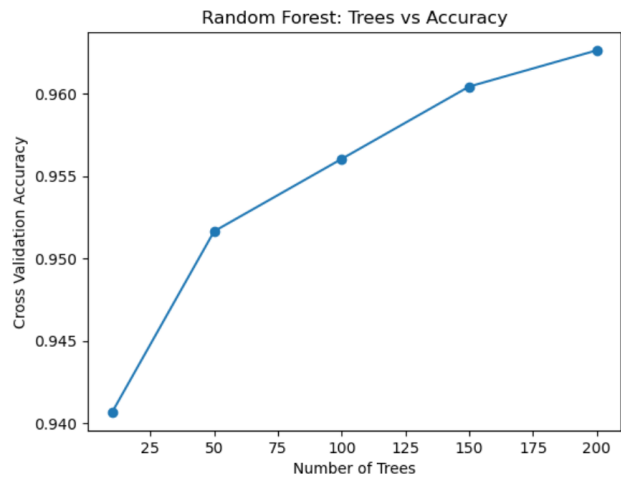


Figure 10: Random Forest: Trees vs Accuracy

Figure 11: Overall figure caption

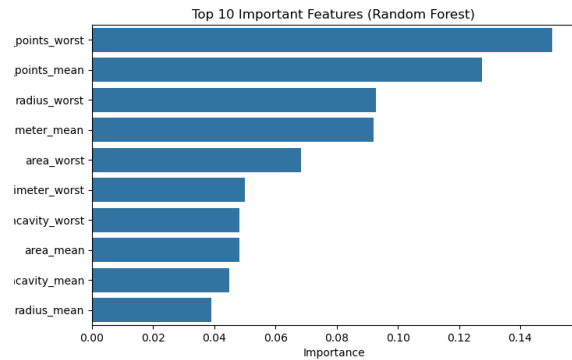
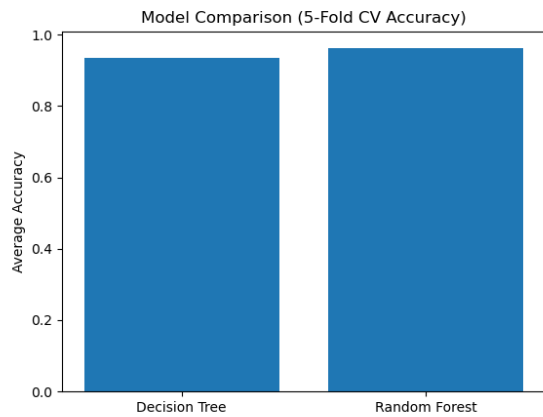


Figure 12: Model Comparison (5-Fold CV Accuracy)

Figure 13: Top 10 Important Features (Random Forest)

Figure 14: Overall figure caption

Hyperparameter Tuning Results

Decision Tree Cross-Fold Results

Table 1: Decision Tree Hyperparameter Evaluation using 5-Fold Cross-Validation

Criterion	Max Depth	Avg CV Accuracy (%)	Avg CV F1 Score
gini	3.0	92.75	90.07
gini	5.0	92.53	89.87
gini	7.0	91.65	88.78
entropy	3.0	93.41	91.11
entropy	5.0	94.73	92.82
entropy	7.0	93.41	91.26

Random Forest Cross-Fold Results

Table 2: Random Forest Hyperparameter Evaluation using 5-Fold Cross-Validation

n_estimators	Max Depth	Max Features	Avg CV Accuracy (%)	Avg CV F1 Score
50	5.0	sqrt	95.1648	93.2726
50	5.0	log2	94.9451	93.0383
50	10.0	sqrt	95.3846	93.6470
50	10.0	log2	94.7253	92.7503
50	NaN	sqrt	95.3846	93.6470
50	NaN	log2	94.7253	92.7503
100	5.0	sqrt	94.9451	92.9841
100	5.0	log2	95.1648	93.3461
100	10.0	sqrt	95.3846	93.5995
100	10.0	log2	94.9451	93.0944
100	NaN	sqrt	95.3846	93.5995
100	NaN	log2	94.9451	93.0944
150	5.0	sqrt	94.9451	92.9841
150	5.0	log2	95.3846	93.6826
150	10.0	sqrt	95.1648	93.3206
150	10.0	log2	95.3846	93.6805
150	NaN	sqrt	95.1648	93.3206
150	NaN	log2	95.3846	93.6805

5-Fold Cross-Validation Performance Comparison

Table 3: 5-Fold Cross-Validation Accuracy Comparison

Model	Fold 1	Fold 2	Fold 3	Fold 4	Fold 5	Average
Decision Tree	95.60	91.21	97.80	94.51	94.51	
Random Forest	95.60	94.51	97.80	94.51	94.51	

Conclusion

Decision Tree and Random Forest models were implemented and evaluated using 5-fold cross-validation. Hyperparameters were selected based on average cross-validation performance, ensuring robust generalization. The results demonstrate that Random Forest reduces variance and improves stability compared to a single Decision Tree.

References

- Scikit-learn: Decision Trees
- Scikit-learn: Random Forest
- UCI Dataset